Teddy Curtis

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Education

PhD, Particle Physics *2021-2026*
Imperial College London
CERN CMS Experiment

MPhys, Theoretical Physics *2016-2020*
First-Class Honours (84%)

Skills

Languages:	Python, SQL, Bash, LaTeX
Machine Learning:	PyTorch, CNNs, RNNs, GNNs, Transformers, RL, Diffusion, NLP/LLM, HuggingFace
Infrastructure:	Distributed Computing, Batch Processing, HPC, Git

Experience

Machine Learning Research Engineer
The Machine Learning Institute / Part-Time *March 2026 – Present*

- **Researching modern ML** with the founder — including VAEs, diffusion and RL — focusing on technical understanding and implementations to guide their advanced industry training programs.

Machine Learning Consultant
LUNRStudio / Part-Time *March 2026 – Present*

- **Designing robust agentic systems** for enterprise integration, building production-ready solutions with strict reliability requirements across diverse business applications.

Doctoral Researcher at CERN/Imperial
CERN/Imperial *Oct. 2021 – Jan. 2026*

- **Analysis lead (8 people), engineered distributed data pipelines** that orchestrate thousands of parallel HPC jobs to clean and process 100+ TB of data.
- **Designed deep learning models** (GNNs, transformers, feedforward networks) for rare-event detection in extremely noisy environments, ensuring robustness under distribution shift.
- **Achieved 30% improvement in signal efficiency with GNN** over classical ML baselines while eliminating manual feature engineering.
- **Published results in leading high-energy physics venues**: EPJC, JHEP (accepted) and a presentation at the Moriond conference — achieving world-leading dark matter constraints.

Technical Lead - Cardiology Researcher
University of Sydney *May 2020 – Jan. 2022*

- **Led development of a patented algorithm** reconstructing full 12-lead ECG from sequential single-electrode measurements, collaborating with cardiologists to validate clinical-grade accuracy.

Masters Student - Theoretical Physics
University of Manchester *Sep. 2019 – June 2020*

- **Achieved state-of-the-art performance** using deep learning methods (CNNs/RNNs/hybrid architectures) for cardiac rhythm classification for Master's thesis.

Technical Projects and Writing

Independent ML research projects and technical blog: https://teddy-curtis.github.io/

- **LLM Speculative Decoding**: Implemented and analysed speculative decoding using Hugging Face, studying how draft-target alignment and rejection sampling affect generation efficiency.
- **Reinforcement Learning for Connect-4**: Developed a DQN training pipeline featuring a transformer-based Q-network, experience replay, and ε-greedy exploration.